Linear Algebra (MT-1004)

Homework#03

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Question 1

Let $A = \begin{bmatrix} 1 & 3 & 4 \\ -4 & 2 & -6 \\ -3 & -2 & -7 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$. Is the equation $A\mathbf{x} = \mathbf{b}$ consistent for all possible b_1, b_2, b_3?

Question 2

Given the following matrix A, use Python's `sympy` library to perform the specified slicing operations:

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$$

Instructions:

1. Define matrix A using `sympy.Matrix`.
2. Extract and print:
   - The second row of the matrix.
   - The third column of the matrix.
   - The 2×2 submatrix from the bottom-right corner.
   - All elements from rows 2 to 4 and columns 1 to 3 (inclusive of row/column indices in 0-based indexing).